

Lab 08: Test and Debug the API

Maps to: A5, A6, A7

Build a pytest regression suite, reproduce a seeded defect and verify the fix.

AI Vibe Coding Assets

- ai_vibe.py - OpenAI Python SDK runner
- mock-data.json - synthetic request and test context
- mock-database.sqlite - inspectable synthetic SQLite seed database
- Prompts.pdf - first-run and refinement prompts
- working-files/generated/ - isolated AI output for review

First-Run Prompt

Using the seeded failing cases in mock-data.json, produce a focused pytest regression suite and a debug plan. Reproduce the failure with the smallest request, assert response and persisted state, identify the likely boundary, and propose the minimum patch without weakening existing acceptance tests.

```
python ai_vibe.py --dry-run
export OPENAI_API_KEY="..."
export OPENAI_MODEL="gpt-5-mini"
python ai_vibe.py
pytest -q
```

Detailed procedure

Step 1: Read Prompts.pdf and identify the role, context, constraints, target files and verification checks.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 2: Inspect mock-data.json; remove any personal, confidential or live credential data before prompting.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 3: Run python ai_vibe.py --dry-run to validate and preview the exact SDK request without using API credits.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 4: Set OPENAI_API_KEY in the shell and choose an available OPENAI_MODEL; never paste either value into source files.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 5: Run python ai_vibe.py and inspect the typed CodeBundle written under working-files/generated/.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 6: Review the generated diff and copy only accepted changes into the working application.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 7: Run pytest -q and capture the seeded failing test.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 8: Read the assertion diff, request payload and response body.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 9: Set a breakpoint in the failing service function.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 10: Reproduce with the smallest single request.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 11: Fix the root cause without changing unrelated behaviour.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 12: Add a regression assertion, rerun the full suite and save the output.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 13: Run pytest -q and the lab acceptance checks; refine the prompt with the observed failure evidence when needed.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Step 14: Save the prompt, model name, generated bundle summary and test result as the lab evidence trail.

Perform the action, observe the output, diagnose any mismatch using the request/status/log evidence, and save one evidence item.

Acceptance criteria

The seeded test fails before the fix; the full suite passes after it; the regression assertion remains.